

Hetal Lad

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EDUCATION

George Washington University
Master of Science in Computer Science

Aug 2025 - May 2027
Washington, DC

BMS Institute of Technology
Bachelor of Engineering in Computer Science

Dec 2020 - May 2024
Bengaluru, India

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Go, C#, C++, PowerShell

Backend & APIs: Node.js, FastAPI, REST APIs, JWT, SSE streaming, CI/CD pipelines, Git, GitHub Actions

Cloud & DevOps: AWS (EC2, S3), Azure, Docker, Vercel, Railway, Linux

Data: PostgreSQL, MySQL, MongoDB, SQLite, SQL query optimization

AI & Tools: LangChain, OpenAI API, RAG pipelines, React, Next.js, Claude Code, Cursor, Power BI

PROFESSIONAL EXPERIENCE

Siemens Healthineers

Aug 2024 - Aug 2025

Associate Software Developer | Syngo.via - regulated clinical imaging platform, C# production codebase

Bengaluru, India

- Resolved 50+ production incidents across distributed medical software services; led root-cause analysis and deployed fixes within Scrum sprint cycles, preventing recurrence through improved test coverage.
- Automated hotfix and release deployment pipelines using PowerShell and batch scripting, reducing system downtime by 18% with repeatable, auditable workflows.
- Built and maintained automated regression test suites in C# across four cross-functional Agile teams, raising coverage to 90% and reducing post-release defects in clinical environments.
- Extended medical imaging software with multi-language support, implementing localization across C# UI components and configuration layers.

Siemens Healthineers

Jan 2024 - Aug 2024

Software Developer Intern

Bengaluru, India

- Developed PowerShell scripts automating backend configuration and client environment setup, cutting onboarding time by 30% and eliminating manual handoff steps across deployment workflows.
- Extended automated test suites in C# supporting cross-functional Agile teams, contributing to regression coverage improvements across the product release cycle.

Kosha.AI

Sep 2023 - Oct 2023

Software Developer Intern

Bengaluru, India

- Customized ERPNext modules using the Frappe framework in Python, reducing API response latency by 25% through targeted database query optimization.
- Implemented modular automation scripts to streamline internal ERP workflows, reducing manual steps and improving reliability of recurring business processes.

PROJECTS

Java IDE Plugin | Eclipse JDT, AST Parsing, PlantUML, SVG, Java 11

2026

Scrum Master & Developer

GWU

- Designed and built an Eclipse IDE plugin that parses Java codebases using Eclipse JDT and AST analysis to automatically generate interactive PlantUML class diagrams with drag-and-drop SVG rendering - reducing manual documentation effort for complex enterprise Java projects.
- Served as Scrum Master leading sprint planning, standups, and retrospectives; made deliberate architectural tradeoff choosing SVG + vanilla JS over heavier frameworks to meet Java 11 compatibility constraints across enterprise environments.

Cardiology RAG Assistant | Python, LangChain, OpenAI ada-002, Pinecone, GPT-3.5

2026

- Built end-to-end knowledge extraction pipeline over 529 pages of AHA clinical guidelines - chunked to 3,123 segments, embedded, indexed in Pinecone, served via GPT-3.5 with citations; directly applicable to compliance document Q&A and regulatory knowledge bases.
- Identified and fixed 3 production failure modes: junk chunk pollution (similarity 0.86+), missing confidence thresholds, and context window overflow - improving retrieval accuracy and reliability of AI-generated compliance answers.

DevOps Copilot | FastAPI, Next.js, TypeScript, SQLite, llama.cpp, JWT

2026

- Built full-stack AI tool that ingests system log data and returns structured root-cause analysis, diagnostic steps, and remediation recommendations via SSE streaming across 3 async layers with JWT auth and SQLite persistence.
- Self-diagnosed and resolved a critical async session lifecycle bug by instrumenting the pipeline and tracing the failure to its source; owned the entire development and deployment lifecycle independently.